Roomifi – AI-Based 2D Floor Plan Color Enhancement Platform

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1. Project Title

Roomifi: An AI-Powered 2D Floor Plan Color Enhancement and Visualization System Using Puter.js

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2. Introduction

Architects, interior designers, and civil engineering students frequently work with basic black-and-white 2D floor plans. While these plans accurately represent structural layouts, they lack visual clarity, color differentiation, and presentation quality. Manual conversion of such plans into colorful, presentation-ready layouts requires time, expertise, and professional design software.

This project proposes developing a web-based AI-driven system that automatically converts simple 2D floor plans into enhanced, colorful 2D architectural designs within seconds. The system will leverage **Puter.js** for AI rendering, permanent file hosting, and key-value metadata storage, eliminating the need for traditional backend infrastructure.

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3. Problem Statement

Traditional 2D floor plans present several challenges:

- Monochrome and difficult to interpret visually
- Time-consuming to enhance manually
- Require specialized software for presentation
- Lack of integrated project management and storage

There is a need for an automated, scalable, and intelligent system that:

- Enhances visual clarity automatically
- Adds meaningful color differentiation
- Preserves architectural structure
- Stores project history with persistent access
- Provides secure user-based data management

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4. Project Objectives

The primary objectives of this project are:

1. Develop a Single Page Application (SPA) using **React + Vite + React Router v7**
2. Integrate **Puter.js AI services** to transform black-and-white 2D floor plans into colorful 2D designs
3. Implement authentication and user-based project management
4. Store and retrieve project metadata using Puter key-value storage
5. Provide a side-by-side comparison tool between the original and enhanced images
6. Deploy the system for public access

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5. Proposed System Overview

The proposed system will:

- Allow users to register and authenticate
- Upload 2D floor plan images via drag-and-drop
- Store images using Puter permanent hosting
- Save metadata (user ID, visibility, URLs) in Puter KV storage
- Process images using Puter AI rendering services
- Generate enhanced, colorful 2D architectural designs
- Allow users to toggle project visibility (Public/Private)
- Display user project history in a dynamic gallery
- Provide a side-by-side comparison between the original and AI-enhanced images

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6. Technology Stack

Frontend

- React
- Vite
- React Router v7
- Tailwind CSS

Backend & Cloud Services (Using Puter.js)

- Puter.js AI Rendering Service
- Puter Serverless Workers
- Puter Permanent File Hosting

- Puter Key-Value Storage

Development Tools

- Git for version control
- AI-assisted code review tools
- Modern development workflow

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7. System Architecture

High-Level Architecture

User
↓
React Frontend (Vite + React Router v7)
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Puter.js API Layer
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AI Rendering Engine (Puter AI)
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File Hosting & KV Storage (Puter)
↓
Frontend Display

Main Modules

- Authentication Module
- Upload Module
- AI Processing Module
- Project Management Module
- Comparison Viewer Module

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8. Key Features

- User Registration & Login
- Drag-and-Drop Floor Plan Upload
- AI-Based 2D Color Enhancement
- Persistent Image Hosting with Public URLs
- Dynamic Project Gallery
- Public/Private Project Toggle
- Side-by-Side Image Comparison Viewer
- Fully Responsive Modern UI

9. 4-Week Development Timeline

Week 1 – Planning & Frontend Setup

- Requirement analysis
- System architecture design
- Initialize project using React + Vite
- Configure React Router v7
- Design UI structure (Navbar, Layout, Homepage)
- Implement the authentication interface

Deliverable:
Working on the frontend with routing and authentication structure.

Week 2 – File Upload & Project Management

- Implement drag-and-drop image upload
- Integrate Puter file hosting
- Implement project creation logic
- Store metadata in Puter KV storage
- Develop project gallery UI
- Implement public/private visibility toggle

Deliverable:
Users can upload floor plans and manage stored projects.

Week 3 – AI Integration & Core Processing

- Integrate Puter AI rendering API
- Design prompt for 2D color enhancement
- Connect frontend to AI service
- Store enhanced image URLs
- Implement loading states and error handling

Deliverable:
Fully functional AI-powered 2D enhancement system.

Week 4 – Comparison Tool, Testing & Deployment

- Implement a side-by-side image comparison viewer

- Optimize UI responsiveness
- Functional and usability testing
- Bug fixing and performance improvements
- Final deployment
- Documentation and presentation preparation

Deliverable:
Fully deployed and documented system.

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10. Expected Outcomes

- A functional AI-powered 2D visualization platform
- Faster and automated floor plan enhancement
- Improved presentation-ready architectural designs
- Demonstration of practical AI integration in web applications
- Real-world implementation of full-stack development concepts

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11. Future Improvements

- 2D to 3D transformation feature
- Multiple design and color themes
- PDF export functionality
- Community sharing feed
- Advanced AI customization options
- Role-based access control

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12. Conclusion

Roomifi demonstrates the practical integration of artificial intelligence with modern web technologies using a serverless architecture powered by Puter.js. The project eliminates traditional backend complexity while delivering scalable AI processing, secure data storage, and persistent hosting.

This project showcases competencies in:

- React-based SPA development
- Client-side routing and state management
- AI service integration
- Cloud-based file hosting
- Serverless architecture
- Full-stack system design principles